

# Which vendors concentrate the vulnerabilities already exploited in real attacks?

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An AWS data lake, 100% Terraform, that answers this question with public CISA data — and measures what every answer costs.

**Group g07** · CESAR School · 2026.2

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# Why the KEV catalog: the real-attack filter

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- **KEV** (*Known Exploited Vulnerabilities*) is the official catalog of **CISA**, the US cybersecurity agency: a CVE only gets in with **confirmed exploitation**.
- It carries weight: US federal agencies have a **mandatory deadline** to fix whatever enters it (directive BOD 22-01).
- **Real, live data**: public, free, and updated every week. It grew from 1,713 to **1,734** in 18 days.

**Our question:** which vendors concentrate the exploited vulnerabilities of the last 12 months — and what does it cost to answer that?

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**393,026**

CVEs published worldwide (NVD)

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**1,713 (0.4%)**

with confirmed exploitation (KEV)

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**299**

added in the last 12 months

# Architecture: everything provisioned as code

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1

## Collect

Downloads the public CISA feed.

ingest.sh

2

## Keep

> Original copy, byte for byte: proof of origin.

S3 · raw layer

3

## Organize

> Cleans and standardizes: one row per CVE.

S3 · trusted + Glue Catalog

4

## Ask

> SQL with a spending cap per query.

Athena WorkGroup (100 MB)

**Terraform** packaged as a module · remote state in **S3 + DynamoDB** · workspace **av1** · schema declared in code, **no Glue Crawler**.

# Trustworthy data before any analysis

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## GRAIN

0

duplicate CVEs: **1,713 rows, 1,713 distinct CVEs**. One row per CVE, proven by query.

- Per-product grain rejected: **214** of 1,713 (12.5%) have an ambiguous product name.

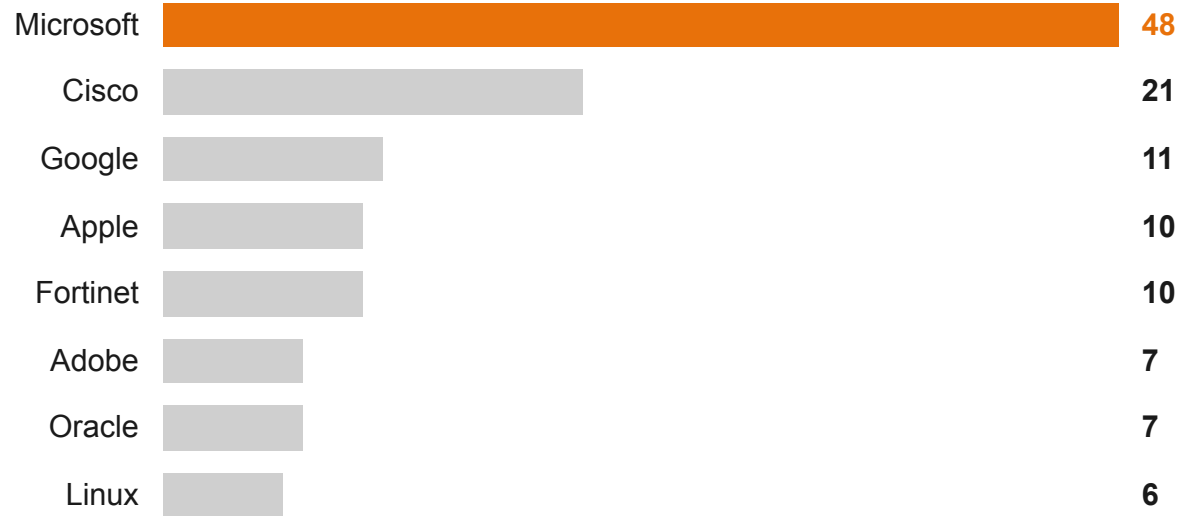
## QUALITY RAW → TRUSTED

18 → 0

names with stray whitespace. Without cleaning, the same vendor becomes **two groups** in the ranking.

- **5** quality gates: if one fails, nothing is published.

# The answer: volume is not risk



Exploited CVEs added in the last 12 months, by vendor (299 in total). Microsoft alone accounts for 16%.

## LINKED TO RANSOMWARE

**10% vs 43%**

Microsoft (5 of 48) · Oracle (3 of 7)

Oracle is only 7th by volume, yet almost half of its vulnerabilities have already been used in ransomware.

# What it costs: measured, not estimated

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**USD 0.00004768**

per query, read from Athena's own execution

**≈ 21,000**

queries for USD 1.00

- The query scans **3 MB**; Athena bills a minimum of **10 MB**. We already pay the floor.
- Parquet and partitioning would save **USD 0.00**. They move to Part 2, once volume passes the floor.
- Re-run from scratch on 2026-10-04 with 1,734 CVEs: **same cost**.

# Demo: from zero to destroy

## 2 Athena query with measured cost

Business question -- top vendors by KEV count, last 12 months

| vendor_project | kev_count | affected_products | ransomware_linked | pct_of_window | most_recent_kev |
|----------------|-----------|-------------------|-------------------|---------------|-----------------|
| Microsoft      | 49        | 15                | 5                 | 15.96         | 2026-09-25      |
| Cisco          | 19        | 13                | 2                 | 6.19          | 2026-09-30      |
| Apple          | 11        | 3                 | 0                 | 3.58          | 2026-09-29      |
| Fortinet       | 11        | 6                 | 0                 | 3.58          | 2026-10-01      |
| Google         | 10        | 5                 | 0                 | 3.26          | 2026-09-16      |
| Linux          | 9         | 1                 | 0                 | 2.93          | 2026-09-18      |
| Adobe          | 8         | 6                 | 0                 | 2.61          | 2026-09-24      |
| Oracle         | 7         | 5                 | 3                 | 2.28          | 2026-08-24      |
| Citrix         | 6         | 2                 | 0                 | 1.95          | 2026-10-04      |
| Synacor        | 6         | 2                 | 0                 | 1.95          | 2026-08-21      |
| Ivanti         | 5         | 3                 | 0                 | 1.63          | 2026-06-11      |
| SonicWall      | 5         | 2                 | 2                 | 1.63          | 2026-09-02      |
| Broadcom       | 4         | 4                 | 1                 | 1.3           | 2026-08-18      |
| Check Point    | 4         | 3                 | 1                 | 1.3           | 2026-09-22      |
| JFrog          | 4         | 1                 | 0                 | 1.3           | 2026-09-11      |

### Measured cost

data scanned : 3077872 bytes  
billed : 10485760 bytes (Athena 10 MB minimum applies)  
engine runtime : 807 ms (total 982 ms)  
cost : USD 0.00004768 at USD 5.00/TB  
i.e. : ~20973 runs of this query per USD 1.00

## 1 apply: resources created in workspace av1

```
OK Workspace      : av1
OK Raw bucket     : eda262-g07-lake-raw
OK Trusted bucket : eda262-g07-lake-trusted
OK Glue database  : eda262_g07_kev
OK Trusted table  : kev_vulnerabilities [grain: one row per cve_id]
OK Athena WorkGroup: eda262-g07-wg
```

## 3 verifica.sh: 25 PASSA / 0 FALHA

```
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PASSA Athena WorkGroup (eda262-g07-wg)
PASSA Dados carregados na camada trusted
PASSA Tags obrigatorias aplicadas nos recursos

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Resultado: 25 PASSA / 0 FALHA (total: 25 criterios)
Todos os criterios foram atendidos.
```

## 4 destroy: 27 resources removed, 0 orphans

```
Destroy complete! Resources: 27 destroyed.
==> Verifying no resources were orphaned
OK No orphaned resources remain
```

# Every decision, one number

| DECISION                   | THE NUMBER BEHIND IT      |
|----------------------------|---------------------------|
| One row per CVE            | 0 duplicates              |
| Raw → trusted cleaning     | 18 → 0 dirty rows         |
| Schema in code, no Crawler | USD 0.073 avoided per run |
| No Parquet in Part 1       | USD 0.00 saved            |
| Destroy verified on AWS    | 0 orphaned resources      |

## FROM PROJECT TO PRODUCT

The foundation of a **patch-prioritization** service driven by real risk, at **USD 0.00004768** per query.

## PART 2

- **EPSS**: exploitation probability for **373k** CVEs, every day.
- Partitioning measured: **12.2×** less data read.

**Repository:** [github.com/brunoribeiro/eda262-projeto-g07](https://github.com/brunoribeiro/eda262-projeto-g07)  
**All decisions:** DECISIONS.md